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Original Research

Independent assessment of the WHO Skin Neglected Tropical Diseases application for leprosy detection

Patrícia Deps,[1] Bianca Barros Canhamaque Amorim,[1] Taynah Repsold,[1] Douglas Almonfrey,[2] Rachel Bertolani do Espírito Santo,[1] Rafael Maffei Loureiro[3], Nkechi Anne Enechukwu,[4] Thiago Zanetti Barreiro,[2] Mecciene Mendes Rodrigues,[5] Andrea Maia Fernandes Fonseca,[6] Gleice Nunes Lima,[7] Marcos César Florian[8], Jose Antonio Ruiz-Postigo[9]

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ABSTRACT

Objectives. To independently evaluate the World Health Organization (WHO) Skin Neglected Tropical Diseases (NTDs) application, focusing on the diagnostic performance of its underlying artificial intelligence model for leprosy detection. The primary objective was to determine the proportion of images in which leprosy appeared among the model's Top-5 diagnostic predictions. The secondary objective was to qualitatively analyze diagnostic error patterns.

Methods. A data set of 439 anonymized clinical images from confirmed leprosy cases (1996–2024) was analyzed, spanning the full clinical spectrum (indeterminate, tuberculoid, borderline/dimorphous, and lepromatous/Virchowian forms) and including reactional and atypical presentations. After excluding 16 images due to processing errors, 423 images were retained: 367 classical leprosy lesions and 56 reactional or atypical leprosy-related presentations. All images were evaluated using the WHO desktop version of the visual classifier. Top-5 sensitivity (recall) for leprosy was estimated, alongside a qualitative error analysis focusing on intrapatient inconsistencies and challenging lesion types.

Results. The model achieved an overall Top-5 sensitivity (recall) of 84.9%, with higher sensitivity for classical lesions (87.2%) than for reactional or atypical presentations (69.6%). Qualitative review revealed inconsistent predictions for visually similar lesions from the same patient, and misclassifications concentrated among necrotic, inflammatory, and infiltrative lesions.

Conclusions. The WHO Skin NTDs application demonstrates substantial promise as a clinical decision-support and educational tool, especially for classical leprosy. Performance gaps for reactional and atypical forms highlight the need for algorithmic refinement. Enhancing data set diversity and integrating patient-level context may improve diagnostic robustness.

Keywords Leprosy; neglected diseases; artificial intelligence; decision support systems, clinical; mobile applications; dermatology; deep learning.

[1] Department of Public Health, Post-Graduate Programme of Infectious Diseases, Federal University of Espírito Santo, Vitória, Brazil. ✉ Patrícia Deps, patricia.deps@ufes.br
[2] Department of Electronic Engineering, Federal Institute of Espírito Santo, Vitória, Brazil.
[3] Department of Radiology, Hospital Israelita Albert Einstein, São Paulo, Brazil.
[4] Dermatology and Infectious Disease Unit, Department of Internal Medicine, Nnamdi Azikiwe University, Nnewi Campus, Nnewi, Nigeria.
[5] Department of Dermatology, Federal University of Pernambuco, Campus Agreste, Caruaru, Brazil.
[6] Serviço de Infectologia de Petrolina (SEINPe), Secretaria de Saúde da Prefeitura Municipal de Petrolina, Petrolina, Brazil.
[7] Secretaria Municipal de Saúde de Confresa, Mato Grosso, Brazil.
[8] Department of Dermatology, Paulista Medical School, Federal University of São Paulo, Brazil.
[9] World Health Organization, Global Neglected Tropical Disease Programme, Skin NTD, Geneva, Switzerland.

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Leprosy (Hansen's disease), a neglected tropical disease (NTD), remains a major global public health concern. In 2024, 172 717 new leprosy cases were reported from 188 countries and territories, with the highest burden in low- and middle-income regions (1). Diagnosis is often complex due to the wide clinical spectrum, which ranges from subtle hypochromic macules in paucibacillary forms to diffuse nodular or infiltrative lesions in multibacillary presentations. Reactional states—immune-mediated inflammatory episodes that may occur before, during, or after treatment—add further diagnostic complexity and contribute to misclassification and delayed management (2).

Limited specialist availability in endemic regions contributes to substantial diagnostic delays, particularly in remote or resource-constrained settings where frontline providers frequently have limited training in recognizing early or atypical manifestations of leprosy (3, 4). These structural gaps are often compounded by persistent stigma, fragmented referral pathways, and operational constraints within primary health services, all of which hinder timely case identification. Consequently, many patients are diagnosed only after the onset of nerve damage and visible deformities, increasing the risk of long-term disability and perpetuating transmission in affected communities (5).

Digital decision-support tools, including image-based artificial intelligence (AI), may help narrow these gaps by assisting nonspecialists in triaging skin lesions and prioritizing differential diagnoses (6, 7). In dermatology, AI systems have shown high concordance with expert assessment in controlled evaluations; however, reports of performance comparable to or exceeding clinicians generally relate to narrowly defined and highly controlled image-classification tasks. The extent to which such performance generalizes to routine clinical practice, especially for clinically heterogeneous diseases, remains uncertain (8–13).

For leprosy, early studies have demonstrated the feasibility of lesion recognition from images, and multimodal approaches combining images with clinical or demographic features may improve diagnostic robustness (14–16). Nevertheless, much of the existing evidence is limited by sample size, restricted image diversity, and a lack of independent external analytical validation.

To address case management challenges in frontline settings, the World Health Organization (WHO) developed the Skin NTDs mobile application (app), a tool that combines offline clinical algorithms with structured learning resources for health workers. The app covers 12 skin NTDs—leprosy, Buruli ulcer, yaws, cutaneous leishmaniasis, post-kala-azar dermal leishmaniasis, onchocerciasis, lymphatic filariasis, mycetoma, chromoblastomycosis, sporotrichosis, scabies, and tungiasis—as well as 24 common skin conditions (17). In evaluations conducted in Ghana and Kenya, the app achieved a mean quality score of 4.02 (standard deviation 0.47) out of 5, supporting its perceived usefulness as a training and decision-support tool (18, 19).

A beta version of the WHO Skin NTDs app incorporates two cascaded AI-based visual classifiers, developed in collaboration with UniversalDoctor and Belle.ai (20). Both models are built on convolutional neural network architectures. The first classifier acts as a broad skin-condition screener, assigning probabilities across 24 common dermatological conditions and indicating when a lesion may warrant further evaluation for a skin-related neglected tropical disease (skin NTD). When a potential NTD is suspected, a second classifier is applied; this NTD-focused model assigns probabilities across the aforementioned 12 skin NTD categories and returns a ranked list of four suggested NTD diagnoses for each uploaded image. Together, these AI components enable the app to function as both a clinical decision-support tool and a scalable training platform for nonspecialist health workers, who can capture and upload anonymized images of skin lesions to obtain a shortlist of possible diagnoses to be integrated with clinical history and examination findings (7, 21).

Although the app displays four diagnostic suggestions in routine use, the underlying classifier generates five ranked outputs, which are accessible in the desktop version of the visual classifier and were used for analytical validation in this study. All analyses in this study refer exclusively to the NTD classifier (second-stage model).

To the authors' knowledge, no previous peer-reviewed study has reported an external analytical validation of the AI-based NTD classifier integrated into the WHO Skin NTDs app. Using an independent data set of clinically confirmed leprosy cases and their reactional forms, this study assessed the classifier's ability to identify leprosy among the 12 NTD output labels—focusing on Top-5 sensitivity and error patterns—and evaluated its potential utility as a digital health intervention supporting global efforts to control leprosy.

OBJECTIVES

This study aimed to assess the performance of the WHO Skin NTDs app's AI model in identifying leprosy and its reactional forms. The specific objectives were to assess the model's sensitivity, defined as the proportion of images in which leprosy appeared among the Top-5 diagnostic predictions generated for each image, and to identify and analyze diagnostic error patterns, with an emphasis on reactional states and atypical clinical presentations.

METHODS

Study Design

This was an external validation study designed to assess the diagnostic performance of a pretrained deep learning model currently integrated into the WHO Skin NTDs mobile application. The validation focused on the model's ability to detect leprosy using an independently curated data set that had not been used during training or internal validation, as recommended by the WHO (22).

Data Collection and Data Set

Image Acquisition and Data Security

Clinical images were collected between 1996 and 2024 by leprosy experts. All images originated from patients with confirmed diagnoses of leprosy and represented authentic, real-world clinical scenarios, including a range of lesion types and skin tones.

Strict data protection protocols were applied throughout the study. Before upload, each image was reviewed and de-identified by trained staff using manual cropping and selective

masking to remove potential identifiers (e.g., background elements, tattoos, jewelry, name tags, or embedded labels). When facial regions were clinically relevant, de-identification was performed by limiting the field of view and obscuring nonessential features while preserving the lesion of interest. De-identified images were then stored in a restricted-access environment and uploaded to the WHO internal SharePoint platform; access to images and associated records was restricted to authorized researchers. For model testing, images were provided without labels or metadata, which were stored separately, to ensure unbiased performance evaluation. The sharing of images adhered to a formal Federal University of Espírito Santo (UFES)–WHO institutional agreement that regulates authorized use and ensures compliance with applicable ethical and data-protection standards.

External Validation Data Set

The external data set comprised 439 anonymized clinical images from 208 confirmed leprosy cases. Because some cases contributed more than one image (e.g., multiple lesions or anatomical sites), the unit of analysis was the image, reflecting real-world use in which the app generates one prediction per photograph. Accordingly, performance metrics are reported at the image level; this study additionally explored intrapatient variability qualitatively. Overall, the data set spanned the full clinical spectrum (indeterminate, tuberculoid, borderline/dimorphous, and lepromatous/Virchowian forms), corresponding to the Ridley–Jopling spectrum. For the present analysis, 16 images with processing errors were excluded, resulting in 423 images divided into two subsets:

- Subset 1 (n = 367): Images of classical leprosy lesions, used to assess the model's ability to detect leprosy among its Top-5 predicted diagnoses;
- Subset 2 (n = 56): Images of leprosy reactions (e.g., type 1 reaction, type 2 reaction) and rare or atypical leprosy-related presentations (e.g., Lucio's phenomenon, erythema multiforme–like lesions), evaluated using the same leprosy-detection endpoint, without subtype-specific analysis due to current model limitations.

Demographic and phenotypic metadata were available for 418 of the 423 images (Subset 1: 362/367; Subset 2: 56/56) and are summarized in Table 1. Within these metadata records, some variables had missing values due to legacy documentation; sex and Fitzpatrick phototype were not inferred from photographs and are reported as missing.

Diagnostic Confirmation

Leprosy diagnosis was established according to WHO guidelines (23), requiring at least one of the following cardinal signs:

1. Loss of sensation in a hypopigmented or erythematous skin lesion;
2. Enlargement of peripheral nerves associated with sensory or motor impairment; or
3. Detection of acid-fast bacilli in slit-skin smear microscopy.

In all cases, diagnoses were established by experienced clinicians following WHO criteria and recorded in structured study metadata. Paucibacillary cases were classified clinically (≤5 lesions and a negative smear), while multibacillary cases were defined by >5 lesions, a positive smear (bacterial index ≥1), or the presence of multiple thickened nerves. When clinically indicated, diagnosis could be supported by additional methods such as histopathology, polymerase chain reaction for *M. leprae*, anti-phenolic glycolipid-1 serology, or peripheral nerve ultrasound. However, given the long collection period and legacy curation, case-level documentation of which ancillary method(s) were performed was not consistently available across records; therefore, method-specific frequencies are not reported. Leprosy reactions were classified by experienced clinicians according to WHO guidance (24).

TABLE 1. Demographic and phenotypic characteristics of the external validation subsets

Variable	Subset 1 (n = 362)	Subset 2 (n = 56)
Sex		
Male	56.9% (206)	91.1% (51)
Female	32.6% (118)	8.9% (5)
Missing	10.5% (38)	0% (0)
Age group		
Adults	87.0% (315)	98.2% (55)
Adolescents	11.0% (40)	0% (0)
Children	1.7% (6)	1.8% (1)
Missing	0.3% (1)	0% (0)
Fitzpatrick phototype		
II	1.1% (4)	7.1% (4)
III	52.5% (190)	64.3% (36)
IV	18.5% (67)	8.9% (5)
V	16.6% (60)	0% (0)
VI	11.3% (41)	7.1% (4)
Missing	0% (0)	12.5% (7)

***Note:** Missing values reflect incomplete legacy metadata in a subset of retrospective images; sex and Fitzpatrick phototype were not inferred from photographs when not recorded. Fitzpatrick phototype I was not observed among images with recorded phototype.

Outcome Measures and Evaluation Approach

The primary outcome was Top-5 sensitivity (recall), defined as the proportion of images in which leprosy appeared among the model's five highest-ranked diagnostic predictions produced by the NTD classifier. Top-5 sensitivity was selected because, as a clinical decision-support tool, inclusion of leprosy among the top-ranked differential diagnoses may be sufficient to raise suspicion and prompt appropriate referral or further evaluation. In addition, a Top-N analysis (from Top-1 to Top-5) was conducted to evaluate how often leprosy was prioritized at different ranking levels, thereby providing a more detailed view of the model's diagnostic hierarchy.

The secondary objective was to conduct a qualitative error analysis to further explore the model's limitations. This analysis focused on identifying inconsistencies in prediction patterns across images from the same patient and examining misclassified lesion types. Particular attention was given to reactional forms, necrotic and infiltrative lesions, and scenarios in which visually similar images were inconsistently classified by the model.

Statistical Analysis

All analyses were conducted using R Statistical Software (version 4.2.1; R Foundation for Statistical Computing). Descriptive statistics were reported as absolute frequencies with percentages. Sensitivity was calculated by comparing the algorithm's Top-5 diagnostic predictions against expert-confirmed diagnoses. Subgroup analyses were performed by subset to identify patterns in model performance.

Ethical Compliance

This study was approved by the Research Ethics Committee of the Health Sciences Centre, UFES, under protocol CAAE 83227624.6.0000.5060. The requirement for written informed consent was waived owing to the retrospective nature of the study and the use of fully de-identified data. The research was conducted under a formal collaboration agreement between the WHO Global Neglected Tropical Diseases Programme and UFES, adhering to international ethical standards for research involving human subjects.

RESULTS

Quantitative Analysis

All performance analyses refer exclusively to the NTD classifier (second-stage model). A total of 439 clinical images from 208 clinically confirmed leprosy cases were initially included in the external data set; after excluding 16 images due to processing errors, 423 images were analyzed, comprising 367 classical leprosy lesions (Subset 1) and 56 reactional or atypical leprosy-related presentations (Subset 2). Overall, leprosy was identified within the Top-5 predictions in 84.9% (359/423) of images. Sensitivity was 87.2% (320/367) for classical presentations and 69.6% (39/56) for reactional or atypical presentations, as summarized in Table 2.

The Top-N analysis revealed that leprosy was ranked first in 46.9% of classical images and 30.4% of reactional or atypical images, highlighting performance disparities between subsets. Table 3 presents the proportion of images in which leprosy appeared at each rank position within the model's five most probable diagnostic outputs.

Qualitative Error Patterns and Intrapatient Inconsistencies

In multiple cases, images from the same patient produced inconsistent diagnostic outputs. For example, hypopigmented macules from a single patient were classified both with and without leprosy among the Top-5 predictions, depending on the anatomical location (Figure 1A). Similarly, a patient with typical facial infiltration and madarosis was misclassified, with leprosy absent from the model's Top-5 predictions (Figure 1B). Nodular and infiltrative lesions—especially those with necrotic features consistent with necrotic erythema nodosum leprosum—were often misclassified, with leprosy not appearing among the Top-5 suggested diagnoses (Figure 2A). Hyperpigmented lesions, on the other hand, showed higher rates of correct identification (Figure 2B). These inconsistencies suggest difficulties in handling intrapatient variability and in recognizing reactional or complex inflammatory patterns.

TABLE 2. Sensitivity (recall) of the AI model for each subset

Metric	Subset 1	Subset 2	Combined total
Sensitivity (recall)	87.2%	69.6%	84.9%

TABLE 3. Sensitivity (recall) of the AI model within Top-N algorithm predictions

Data set	Top-1	Top-2	Top-3	Top-4	Top-5
Subset 1	46.9%	63.5%	73.8%	80.1%	87.2%
Subset 2	30.4%	46.4%	58.9%	62.5%	69.6%

FIGURE 1. (A) Hypopigmented macule. (B) Facial infiltration with madarosis. Both images were incorrectly classified by the model, with "leprosy" absent from the Top-5 NTD predictions.

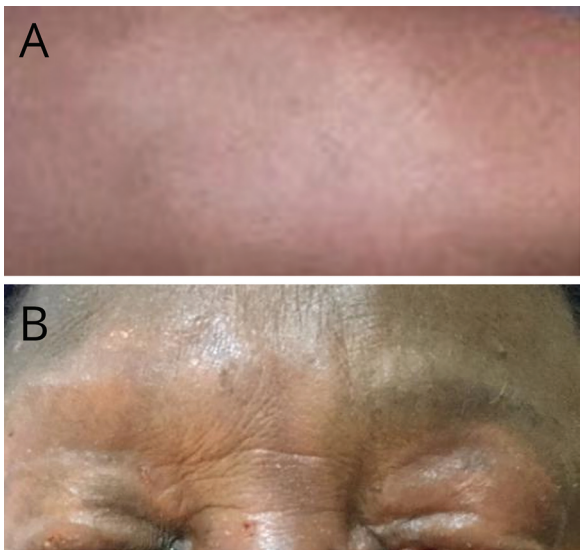
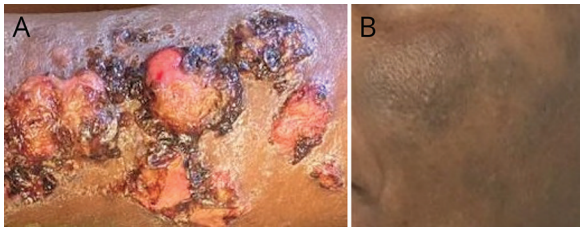


FIGURE 2. (A) Necrotic lesion consistent with necrotic erythema nodosum leprosum, with "leprosy" absent from the Top-5 NTD predictions. (B) Hyperpigmented plaque correctly identified as leprosy within the model's Top-5 predictions.



DISCUSSION

This is the first article to report an independent external analytical validation of the WHO Skin NTDs app's AI model for leprosy diagnosis. Using a large and diverse data set of clinically confirmed cases collected over nearly three decades, the study evaluated the model's ability to recognize both classical leprosy lesions and reactional forms across a broad

spectrum of clinical presentations. A key strength of this work lies in the comprehensiveness of the external data set, which includes images from multiple anatomical sites and spans a broad range of Fitzpatrick skin phototypes among images with recorded phototype. This diversity enhances the generalizability of the findings and aligns with WHO regulatory expectations for external validation, which requires performance assessment using data sets entirely independent of model development (22).

Overall, the AI model demonstrated high sensitivity, identifying leprosy within the Top-5 predictions in 84.9% of images. As expected, performance was stronger for classical lesions (87.2%) than for reactional or atypical presentations (69.6%), which are inherently more difficult to diagnose due to their variable and often inflammatory nature. This discrepancy underscores the need for targeted refinement of the model to better capture immune-mediated or atypical forms. Differences in subset composition may also have influenced performance estimates; Subset 2 consisted predominantly of adults with intermediate skin phototypes and displayed a more uniform phenotypic profile, whereas Subset 1 included a broader distribution of ages and pigmentation patterns. The Top-N analysis reinforced this performance gap, with leprosy ranking first in nearly half of classical lesions but in only one-third of reactional or atypical lesions. Future work should assess whether performance differences across leprosy subtypes and reactional states reflect their relative representation during model development, which may help explain reduced sensitivity in rarer or more heterogeneous presentations.

A recent abstract by Madhumita (25) presented a preliminary evaluation of the WHO Skin NTDs app's AI algorithm using 300 leprosy cases, reporting higher sensitivity for polar forms and reduced accuracy in borderline and reactional states. The current findings extend this early evidence by providing a more robust external validation with a larger data set and by incorporating a detailed qualitative assessment of error patterns.

Beyond aggregate performance metrics, qualitative analysis revealed inconsistencies in predictions for visually similar lesions from the same patient. These inconsistencies may stem from limited intrapatient variability within the training data or from challenges in capturing spatial context. Necrotic, erythematous, and infiltrative lesions were particularly prone to misclassification, reinforcing known limitations of convolutional models in recognizing complex inflammatory patterns (26). To address these issues, future development should prioritize data set enrichment with underrepresented reactional forms and rare phenotypes, which remain poorly represented in most existing AI pipelines for leprosy and other skin NTDs (15, 16). In addition, approaches such as lesion-focused segmentation and the incorporation of patient-level or multimodal context may improve robustness in visually complex presentations and reduce intrapatient inconsistencies (9, 10, 27–29).

The practical relevance of this app is considerable, particularly in primary-care and low-resource settings where access to dermatologists or leprosy specialists is scarce. In such contexts, AI-generated ranked differentials, even when leprosy is not listed first, can facilitate case suspicion and prompt referral (6, 7). The app may therefore function as an accessible decision-support and educational tool for frontline health workers. However, because recognizing reactional states and infiltrative presentations remains clinically critical and may require additional expertise, the app should complement clinical assessment rather than function as a standalone diagnostic tool. Image-based outputs should be interpreted alongside clinical history and examination; structured tools such as the Leprosy Suspicion Questionnaire (LSQ) can further strengthen assessment and referral decisions, particularly for subtle or atypical presentations (30). Given that diagnostic accuracy of primary-care clinicians for common skin diseases ranges from 19% to 36% (21, 31), accessible tools that support triage and referral can offer meaningful incremental benefit.

Considering that leprosy is far less common and often unfamiliar to general practitioners, a tool achieving 87.2% sensitivity for classical presentations—corresponding to 80.1% sensitivity when restricted to the four diagnoses displayed in the app—represents a meaningful adjunct to clinical care. Nevertheless, clinical adoption remains limited by the scarcity of robust external evaluations. Inconsistent reporting and poor transparency further restrict trust and interpretability, highlighting the need for standardized evaluation frameworks tailored to AI in medical imaging (12, 22, 32).

This study has several limitations. First, the data set included only confirmed leprosy cases, preventing the calculation of specificity or false-positive rates. Second, because this was not a prospective implementation study, the authors did not assess real-world workflow integration, user behavior, clinical impact, or field performance. In addition, testing relied on the desktop classifier rather than the mobile app. Third, this study did not include direct comparisons with human raters, which are essential for contextualizing model performance and informing clinical deployment strategies (12, 22). Fourth, because images were collected over a long period (1996–2024), acquisition conditions and camera technology likely varied over time, which may have introduced additional heterogeneity and noise, while also reflecting real-world variability relevant to external validation. Finally, most images depicted isolated lesions without accompanying metadata, limiting the ability to emulate real-world diagnostic reasoning (15, 30), and documentation of ancillary confirmatory tests (performed when clinically indicated) was incomplete in legacy records, preventing reliable estimates of their frequency and precluding stratified analyses by confirmation pathway.

In conclusion, the WHO Skin NTDs app demonstrates considerable promise as a clinical decision-support tool for leprosy, especially in endemic and resource-limited settings. This study confirms its good performance in detecting classical leprosy lesions while identifying persistent gaps in the recognition of reactional and atypical forms. To improve its clinical utility and reliability, the authors recommend the following: (i) developing external evaluation data sets that include a broad range of nonleprosy and look-alike conditions to enable robust estimation of specificity and false-positive rates; (ii) increasing the representation of reactional and rare phenotypes in future evaluations and model refinement; and (iii) conducting clinician comparison studies and prospective field evaluations. With continued refinement and responsible integration, AI-powered digital tools such as the WHO Skin NTDs app can strengthen early detection, reduce disability, and accelerate progress towards global elimination targets.

Author contributions. PD and JAR-P conceived the study. PD, DA, TZB, and JAR-P designed the methodology. PD, BBCA, and RML analyzed the data. PD, BBCA, TR, RBES, NAE, MMR, AMFF, GNL, MCF, and JAR-P curated the data. PD and RML drafted the manuscript. All authors reviewed and approved the final version.

Data availability statement. The data supporting the findings of this study are available from the corresponding author upon reasonable request.

Conflicts of interest. Dr Ruiz-Postigo is affiliated with the World Health Organization; this affiliation did not influence the study design, data analysis, or interpretation of the results. The other authors declare no conflicts of interest.

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Declaration of artificial intelligence (AI) use. The authors used a large language model (ChatGPT, OpenAI) to assist with language editing and text polishing. All content, analyses, and conclusions were independently verified by the authors, who take full responsibility for the integrity of the work.

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Evaluación independiente de la aplicación móvil de la OMS dirigida a las enfermedades cutáneas tropicales desatendidas para detectar la lepra

RESUMEN

Objetivos. Evaluar de forma independiente la aplicación móvil de la Organización Mundial de la Salud (OMS) dirigida a las enfermedades cutáneas tropicales desatendidas (Skin NTDs App), centrándose en el desempeño diagnóstico de su modelo de inteligencia artificial subyacente en la detección de la lepra. El objetivo principal fue determinar la proporción de imágenes en las que la lepra aparecía entre las cinco mejores predicciones diagnósticas del modelo. El objetivo secundario fue analizar cualitativamente los patrones de error diagnóstico.

Métodos. Se analizó un conjunto de datos de 439 imágenes clínicas anonimizadas de casos confirmados de lepra (1996-2024), que abarcaban todo el espectro clínico (formas indeterminada, tuberculoide, limítrofe o dimórfica, y lepromatosa o virchowiana) e incluían presentaciones reactivas y atípicas. Tras descartar 16 imágenes debido a errores de procesamiento, se conservaron 423 imágenes: 367 lesiones clásicas de lepra y 56 presentaciones reactivas o atípicas relacionadas con la lepra. Todas las imágenes se evaluaron con la versión de escritorio del clasificador visual de la OMS. Se estimó la sensibilidad (capacidad de detección) para la lepra considerando las cinco primeras opciones diagnósticas propuestas por el modelo, junto con un análisis cualitativo de errores centrado en las incongruencias entre lesiones de un mismo paciente y en los tipos de lesiones difíciles de clasificar.

Resultados. El modelo alcanzó una sensibilidad general del 84,9% dentro de las cinco primeras predicciones diagnósticas, y mostró mayor sensibilidad para las lesiones clásicas (87,2%) que para las presentaciones reactivas o atípicas (69,6%). La revisión cualitativa puso de manifiesto incongruencias en las predicciones frente a lesiones visualmente similares de un mismo paciente, y se observó una concentración de errores de clasificación en las lesiones necróticas, inflamatorias e infiltrativas.

Conclusiones. La aplicación Skin NTDs App resulta muy prometedora como herramienta educativa y de apoyo para la toma de decisiones clínicas, sobre todo en el caso de la lepra clásica. Las diferencias de desempeño en el caso de las formas reactivas y atípicas hacen patente la necesidad de perfeccionar el algoritmo. Aumentar la diversidad de los conjuntos de datos e integrar el contexto a nivel del paciente podría mejorar la solidez diagnóstica.

Palabras clave Lepra; enfermedades desatendidas; inteligencia artificial; sistemas de apoyo a decisiones clínicas; aplicaciones móviles; dermatología; aprendizaje profundo.

Avaliação independente do aplicativo Skin Neglected Tropical Diseases da OMS para detecção da hanseníase

RESUMO

Objetivos. Avaliar de modo independente o aplicativo da Organização Mundial da Saúde (OMS) para diagnosticar doenças tropicais negligenciadas relacionadas à pele — Skin Neglected Tropical Diseases ou Skin NTDs —, com foco no desempenho do modelo de inteligência artificial desse aplicativo para a detecção da hanseníase. O objetivo primário foi determinar a proporção de imagens nas quais a hanseníase aparecia entre as cinco primeiras previsões de diagnóstico do modelo. O objetivo secundário foi analisar qualitativamente os padrões de erros de diagnóstico.

Métodos. Analisou-se um conjunto de dados de 439 imagens clínicas anonimizadas de casos confirmados de hanseníase (de 1996 a 2024), abrangendo todo o espectro clínico (formas indeterminada, tuberculoide, limítrofe/dimorfa e lepromatosa/virchowiana), com a inclusão de manifestações reacionais e atípicas. Após a exclusão de 16 imagens devido a erros de processamento, foram mantidas 423 imagens: 367 lesões clássicas de hanseníase e 56 manifestações reacionais ou atípicas relacionadas à hanseníase. Todas as imagens foram avaliadas usando a versão para desktop do classificador visual da OMS. Estimou-se a sensibilidade (recall) dos cinco diagnósticos mais prováveis do aplicativo para a hanseníase e realizou-se uma análise qualitativa de erros, com foco em inconsistências intrapacientes e tipos de lesões desafiadoras.

Resultados. A sensibilidade (recall) global dos cinco diagnósticos mais prováveis do modelo foi de 84,9%, com maior sensibilidade para lesões clássicas (87,2%) do que para manifestações reacionais ou atípicas (69,6%). A análise qualitativa revelou previsões inconsistentes para lesões visualmente semelhantes do mesmo paciente e erros de classificação concentrados em lesões necróticas, inflamatórias e infiltrativas.

Conclusões. O aplicativo Skin NTDs da OMS é bastante promissor como ferramenta educacional e de apoio à decisão clínica, especialmente no caso da hanseníase clássica. As lacunas no desempenho do aplicativo para as formas reacionais e atípicas destacam a necessidade de refinamento do algoritmo. Medidas como aumentar a diversidade do conjunto de dados e integrar o contexto do paciente podem melhorar a robustez do diagnóstico.

Palavras-chave Hanseníase; doenças negligenciadas; inteligência artificial; sistemas de apoio a decisões clínicas; aplicativos móveis; dermatologia; aprendizagem profunda.